MINI PROJECT REPORT

ON

A.U.R.A.

Submitted by

CHRISTOPHER JOSHY (SJC23CS035)

DEON GEORGE (SJC23CS086)

AMAL KURIAN ROYCE (SJC23CS035)

MOHAMMAD BINSHID A(SJC23CS171)

to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

in

Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI

(AUTONOMOUS)

APRIL :: 2025

DECLARATION

We undersigned hereby declare that the mini project report on "**A.U.R.A.[AI Assistive User Reminder Application]**" submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by under supervision of **Prof Divya Sunny**, Assistant Professor, Dept of CSE. This submission represents our ideas in our own words and where ideas and words of others have been included. We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a course for disciplinary action by the institute and/or the University and can also evoke panel action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Name and Signature of Student

CHRISTOPHER JOSHY (SJC23CS035)
DEON GEORGE (SJC23CS086)
AMAL KURIAN ROYCE (SJC23CS035)
MOHAMMED BINSHID A (SJC23CS171)

Place : Choondacherry
Date : 13-04-2026

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI

(AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the report entitled "**A.U.R.A. [AI Assistive User Reminder Application]**" submitted by **CHRISTOPHER JOSHY (UNIVTY REG NO)**, **DEON GEORGE (SJC23CS086)**, **AMAL KURIAN ROYCE (SJC23CS035)**, **MOHAMMAD BINSHID A (UNIVTY REG NO)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the mini project work carried out by them under my guidance and supervision.

Mini Project Guide
Prof.Divya Sunny
Assistant Professor
Department of CSE

Mini Project Coordinator
Prof. Smitha Jacob
Assistant Professor
Department of CSE

Head of the Department
Dr. Joby P P
Professor & Head
Department of CSE

Place : Choondacherry
Date : 04-04-2025

ACKNOWLEDGEMENT

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We are extremely indebted to **Dr. Joby P.P, Professor & Head**, Department of Computer Science and Engineering for his valuable suggestions and encouragement during the course of this project work.

We would like to express our gratitude to our project coordinator **Prof. Smitha Jacob**, Assistant Professor, Department of Computer Science and Engineering for his valuable suggestions and guidelines throughout the period of this project work. His contributions and technical support in preparing this report are greatly acknowledged.

We owe our deep gratitude to our project guide **Prof. Divya Sunny**, Assistant Professor, Department of Computer Science and Engineering who took keen interest in this project and guided us all along, till the completion of this project work by providing all the necessary information for developing a good system.

We are thankful and fortunate enough to get constant encouragement, support and guidance from all the staff members of the department of Computer Science and Engineering, which helped us in successfully completing our mini project work.

CHRISTOPHER JOSHY
DEON GEORGE
AMAL KURIAN ROYCE
MOHAMMED BINSHID A

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Abbreviations

some samples

A.U.R.A.	-	AI Assistive User Remainder App
API	-	Application Programming Interface

Abstract

Dementia is a progressive neurological disorder that causes severe memory loss, leaving patients unable to recognize familiar faces, manage medication schedules, track daily routines, or respond safely during emergencies — placing an enormous burden on both patients and caregivers. Existing assistive solutions remain inaccessible to most due to high cost, technical complexity, and lack of regional language support. A.U.R.A (AI Assistive User Reminder Application) is an intelligent, affordable assistive system developed to address these challenges. The system consists of the AURA Module, a camera-equipped hardware unit currently prototyped using a computer webcam, which performs real-time face detection and recognition using a custom-trained deep learning model achieving above 99.6the unified interface for patients and caregivers. The application features Orito, a conversational AI assistant powered by the Google Gemini API that communicates naturally in Malayalam. Orito enables patients to identify approach- ing individuals by simply asking, query today's medications and scheduled events, set reminders through natural conversation, and automatically dispatch SOS alerts to registered caregivers upon detecting patient distress. The backend is built using FastAPI (Python), with all data — including facial embeddings, medication records, event schedules, care- giver registrations, and SOS logs — stored in MongoDB for its flexible document-oriented schema. A.U.R.A implements a dual-role authentication model. Patients undergo an adaptive on- boarding assessment on first login, allowing the application to optimize its interface and interaction style based on their cognitive condition. Caregivers are authorized exclusively via a Gmail OAuth mechanism pre-registered by the patient, with portal access restricted solely to medication details and incoming SOS alerts to preserve patient privacy. Built entirely on open-source frameworks and cloud AI services, A.U.R.A proves that intelligent, culturally sensitive assistive technology can be developed affordably. Future scope includes

dedicated embedded AURA Module hardware, GPS-based patient tracking, wearable fall detection, and expanded multilingual support, establishing A.U.R.A as a scalable foundation for AI-driven dementia care.

Chapter 1

Introduction

Dementia is one of the most challenging neurodegenerative conditions affecting millions of elderly individuals worldwide. It progressively deteriorates memory, cognitive function, and the ability to perform routine daily tasks. Among those affected, patients in regional communities face an even greater challenge — most available assistive technologies are designed for English-speaking, technically literate users, leaving a significant gap for people who communicate in regional languages such as Malayalam. Families caring for dementia patients in Kerala are forced to maintain near-constant physical presence, as existing solutions fail to address the full spectrum of daily needs these patients face. The rapid advancement of artificial intelligence, mobile computing, and speech processing technologies has opened new possibilities for building intelligent assistive systems that are affordable, accessible, and culturally appropriate. However, these technologies have largely not been brought together in a way that directly serves the dementia care context, particularly for regional language users in India. A.U.R.A — AI Assistive User Reminder Application — is developed in response to this gap. It is a smart, integrated system that combines real-time face recognition, a Malayalam-speaking conversational AI assistant named Orito, medication and event management, a Memory Journal, automatic distress detection, bidirectional location tracking, and caregiver alerting — all within a single mobile application built on React Native, backed by a FastAPI server and MongoDB database, and powered by a custom fine-tuned FaceNet model, Google Gemini API, and Sarvam AI's speech services.

1.1 Problem Statement

Clearly define the research problem or gap that your study addresses.

1.2 Objectives and Scope

Specify the goals of your research and what you aim to achieve. Outline the boundaries and focus of the research.

1.3 Methodology Overview

Briefly mention the approach or techniques used in the study.

Chapter 2

Literature Survey

2.1 Literature Referenced

2.1.1 If the reference is a journal or conference paper

Template for title should be like

First Author Name et al. (year) - Paper or conference title

The body contains Research Objective, Methodology Used, Key Findings, Strengths & Contributions, 6. Limitations & Research Gaps, Relevance to your Project

Example: Smith et al. (2022) - IoT-Based Smart Energy Meter for Real-Time Monitoring The integration of IoT in energy metering has been a growing field of research, aiming to enhance monitoring and efficiency in electricity consumption. Several earlier projects, such as the IoT DC Energy Meter and the GSM Prepaid Energy Meter, attempted to au- tomate energy measurement. However, traditional methods still required manual reading of meters, making the process time-consuming and inefficient.

A recent study introduced an IoT-based Electricity Energy Meter using ESP32 and the Blynk application, providing a real-time, remote monitoring solution[1]. This approach automates electricity consumption tracking and offers a cost-effective and time-efficient alternative to traditional methods.

The project incorporates current and voltage sensors to measure power consumption accurately. The SCT-013 Non-Invasive AC Current Sensor is selected for its capability to measure AC currents up to 100A, while the ZMPT101B AC Voltage Sensor Module

ensures precise voltage readings. The collected data is then transmitted to the Blynk dashboard, allowing users to monitor voltage, current, power, and total units consumed in real-time. Despite its advantages, the IoT-based system has certain limitations. The depen- dency on stable internet connectivity poses challenges in areas with network instability. Additionally, sensor calibration is crucial for ensuring accuracy, making the initial setup process demanding.

Overall, IoT-based smart energy meters present a significant advancement in energy monitoring, enabling automated data collection and remote access. Future research could focus on enhancing data reliability in low-connectivity regions and improving sensor calibration techniques to optimize accuracy.

2.1.2 If the reference is a journal or conference paper2

The body contains Research Objective, Methodology Used, Key Findings , Strengths & Contributions, 6. Limitations & Research Gaps, Relevance to your Project.

Do cite the same, for example [2] , means Using [2] in the body of a text is part of the IEEE citation style, which follows a numbered referencing system. This means that sources are cited using bracketed numbers ([1], [2], [3]), and the full reference details appear in a numbered list at the end of the document(References).

2.2 Existing Solutions

2.2.1 Website- If referred a website

If its a website give title as Website Name (Year) - Title of the Webpage or Article

For Eg: Sensors India (2025) - IoT Sensors for Smart Energy Monitoring

The body should contain objective, methodology, Key Findings, Strengths, Limitations, Relevance to your Project

2.2.2 If refereed a mobile App or an application program

Title template should be like [App Name]([Developer],[Year]) - [Short Description of Purpose]

eg: Blynk (Blynk Inc., 2023) - A Mobile IoT Dashboard for Smart Energy Monitoring

Body contains Objective, Features & Methodology, Limitations and Relevance to your Project

Chapter 3

System Study

3.1 Technologies

3.1.1 Title

Body

3.1.2 Title

Body

3.2 Software Components

Body, can add subheadings

3.3 Hardware part

Chapter 4

System Design

4.1 Activity Diagram

mrntion about your activity diagram

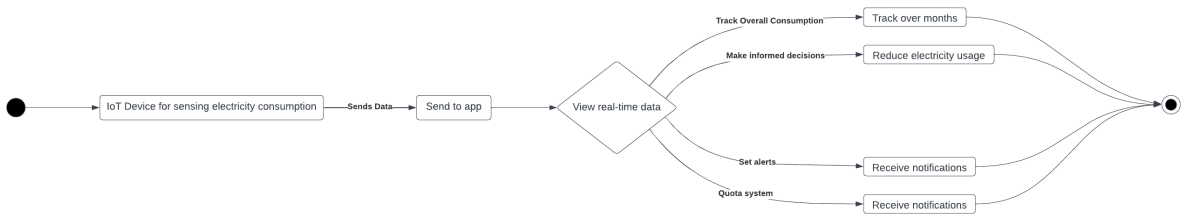

Figure 4.1: Activity diagram

4.2 Sequence Diagram

Explain about diagram

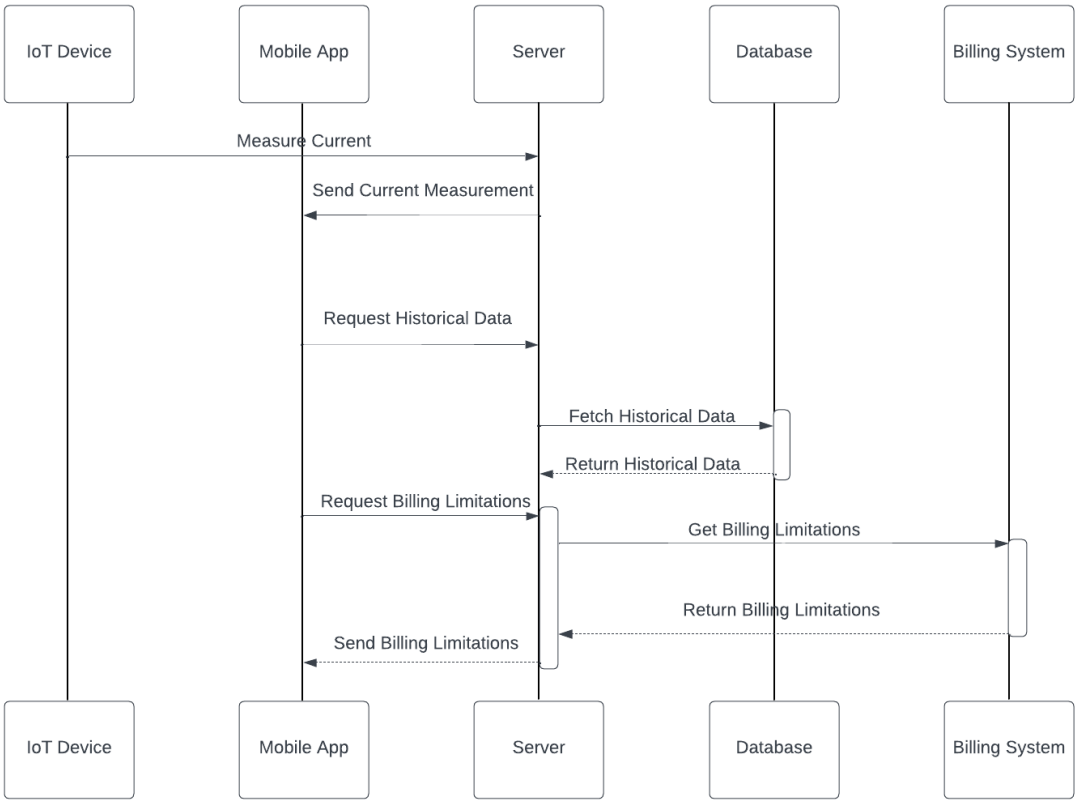

Figure 4.2: Sequence Diagram

4.3 ER Diagram

Explain about diagram.

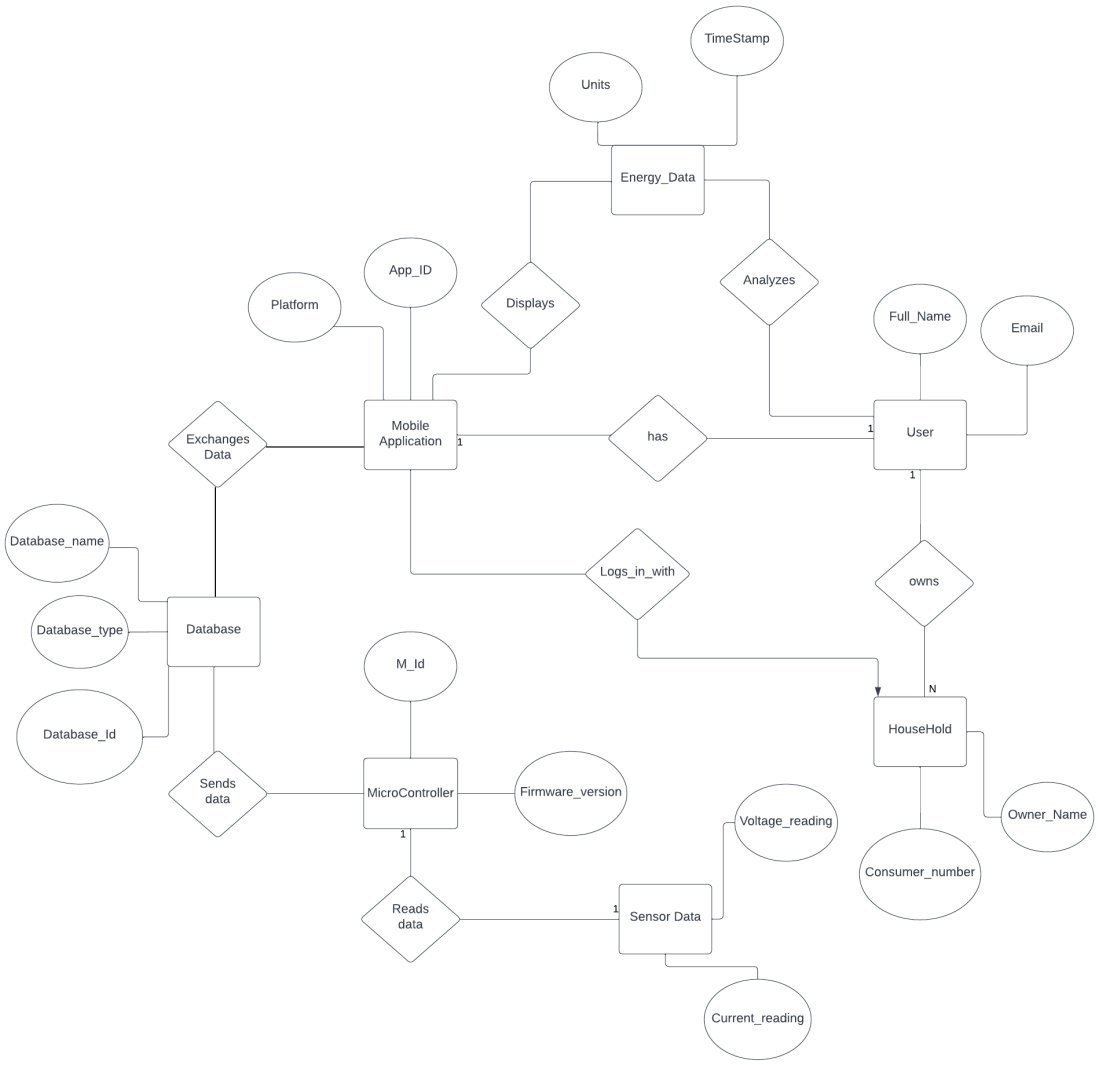

Figure 4.3: ER diagram

4.4 Use Case Diagram

Explain about diagram

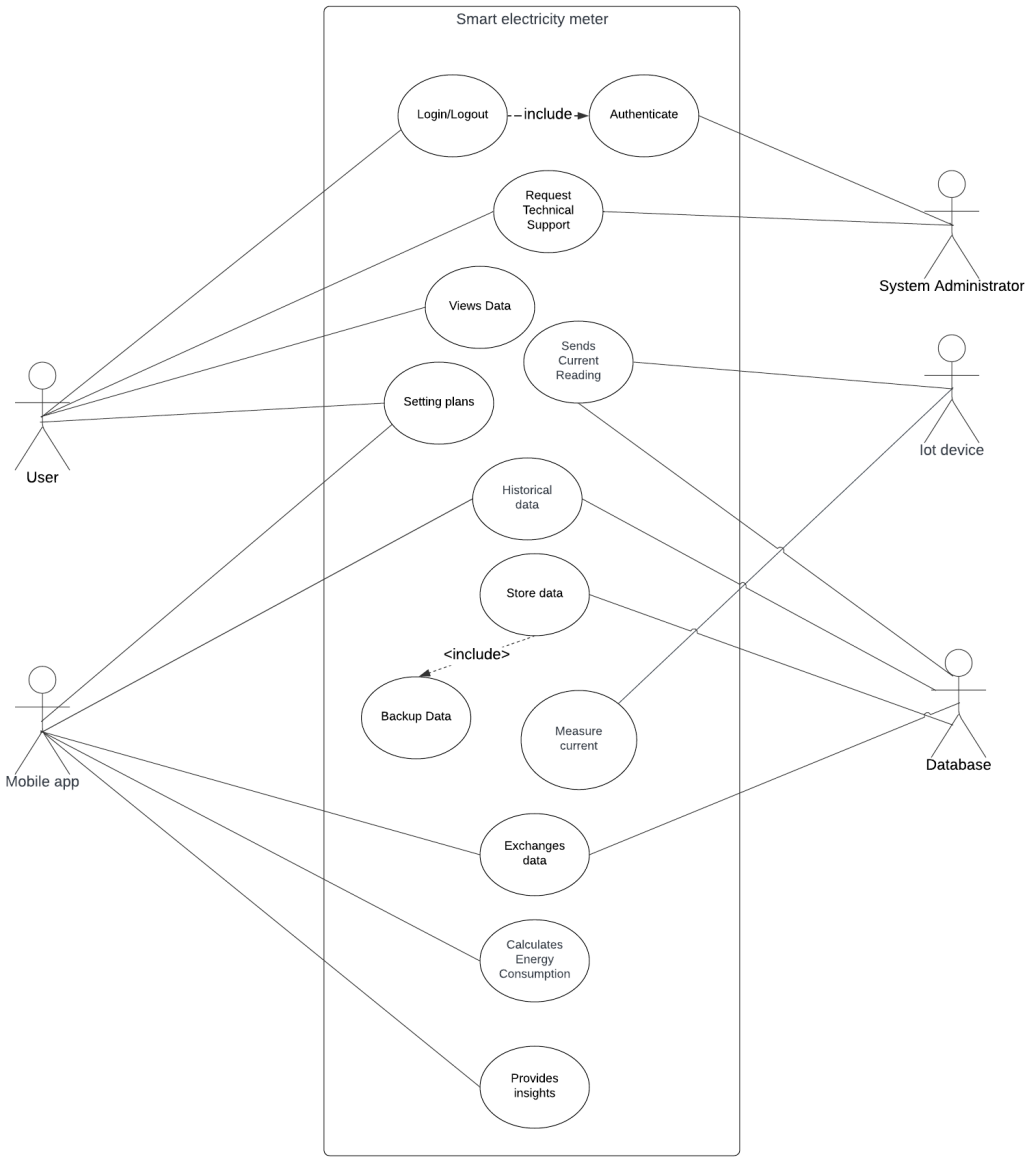

Figure 4.4: Use case diagram

Chapter 5

System Implementation

5.1 Modules

Body

5.1.1 Subheading 1

Body

5.1.2 Subheading 2

Body

Chapter 6

Experimental Results

6.1 Results and Discussions

The system was executed on [mention platform, e.g., Windows/Linux] using [mention software, e.g., Python, Java, MySQL].

The input data, algorithm implementation, and execution process are illustrated in the following screenshots.

6.2 System Execution Output

(Provide screenshots of UI, CLI outputs, API responses, database queries, etc.)

6.2.1 Subtitle-User Panel Execution

Example (for a Web-based System):

Login Page Screenshot

Dashboard Display

Data Processing Output

6.2.2 Subtitle-Admin Panel Execution

Example (for a Web-based System):

Login Page Screenshot

6.2. System Execution Output

Dashboard Display

Data Processing Output

Chapter 7

Conclusion

This project presents a practical application of Artificial Intelligence in healthcare, aimed at supporting individuals with cognitive impairments. The system enhances patient safety, memory assistance, and independence through real-time location tracking, AI-based face recognition. It also includes intelligent reminder modules for daily routines and important events. By automating monitoring and assistance tasks, the system improves patient independence while significantly reducing caregiver workload, forming a strong foundation for future cognitive assistive technologies.

Reference

[1] World Health Organization, "Dementia," Sep. 2023. Accessed: Dec. 29, 2025. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/dementia

[2] Google, "Gemini API Documentation." [Online]. Available: https://ai.google.dev/

[3] SOS Alert System, "Emergency SOS: How it works." [Online]. Available: https://support.google.com/android/answer/9319337

[4] "A comprehensive review of deep learning face recognition methods including FaceNet," Applied Sciences. [Online]. Available: https://www.mdpi.com/2076-3417/15/17/9390

[5] CareYaya's MedaCareLLM system, "AI glasses support dementia patients," Nov. 05, 2024. [Online]. Available: https://healthcare-digital.com/technology-and-ai/careyayas-medacarellm-ai-glasses-support-dementia-patients

[6] MongoDB, "MongoDB Documentation." [Online]. Available: https://www.mongodb.com/

Annexure A

A.U.R.A

PResented bY:

Christopher Joshy
Amal Kurian Royce
Deon George
Mohammad Binshid A

Guide: Prof Divya Sunny

INTRODUCTION

A.U.R.A (AI Assistive User Reminder Application) is an AI-based assistive mobile application designed to help patients.

So what kind of patients ??

TARGET USERS

Memory-impaired Patients

BEFORE THEIR CONDITION

01

- Memory Problems
  - Forgets recent events
  - Repeats questions
  - Has trouble learning new things
- Confusion Issues
  - Gets confused about daily routine
  - Sometimes doesn't recognize people

02

That's why:

- He writes notes to remember his daily events.
- He relies on written reminders to reduce confusion.
- With help of manual taken images they identify people.

EXISTING SYSTEM

CareYaya's MedaCareLLM system = AI + Smart Glasses that:

✔ Recognizes faces and objects in real time
✔ Helps with medications and daily tasks
✔ Collects data to improve dementia research and tracking
✔ Take live conversations convert into text and add to the database, when ever the patient get encounter with the same person the ai give the early conversation as a summary note.

We are actually inspired from the already exiting System and in addition we added some more features also.

WHAT A.I.U.R.A GIVES YOU

01 Daily Routine Reminder System.

02 Face Recognition with Memory Recall.

03 Bot-Based Memory Journal.

04 Emergency SOS Alerts & Location Tracking.

05 Orito-An AI Assitant

MAIN POINT IS A.I.U.R.A. APP RUN ON MOBILE PHONE SO IT IS CHEAPER COMPARE TO SMART GLASSES.

TECH STACK

Frontend (Mobile Application)
- React Native (TypeScript)
- Expo

Authentication
- Firebase Authentication

Backend
- FastAPI (Python)
- Uvicorn

Database & Storage
- MongoDB

TECH STACK

MODELS USED:

- OpenCV → Person / Face Detection
- FaceNet → Face Recognition via Embeddings
- Sarvam Bulbul = TTS
- Sarvam Saaras = STT
- System Output → Draft Memory Bank by Gemini API

LITERATURE REVIEW

STUDY	METHODOLOGY	KEY FINDINGS	LIMITATIONS	FUTURE SCOPE
Deep Learning-based Face Recognition Pretrained model we are using is → **Face Net**	Face images are collected and aligned Deep CNN Model Uses convolutional neural nitworks for feature extraction Each face → 128D vector Triplet Loss Uses: Anchor (A) Positive (P) Negative (N) 👉 Objective: Minimize distance (A, P) Maximize distance (A, N)	Achieved 99.63% accuracy on LFW dataset Reduced error rate by ~30% compared to previous methods Efficient representation: Only 128 bytes per face needed	Sensitive to real-world conditions → affected by occlusion, noise, low resolution. Dataset bias → may give unfair results across race, age, gender. High computational cost → needs large data and GPU	Develop lightweight models for mobile and real-time use Reduce bias for fair and inclusive recognition Enhance accuracy using advanced deep learning techniques

LITERATURE REVIEW

STUDY	METHODOLOGY	KEY FINDINGS	LIMITATIONS	FUTURE SCOPE
Sarvam Bulbul is developed by generate natural, human-like speech for Indian languages, especially handling code-mixed text (like Hinglish). Focus: multilingual + real-world conversational speech	**Clean the text** → Fix and break the text into chunks **Convert text to sounds** → Turn words into how they are pronounced (phonemes) **Create voice pattern** → Generate a sound pattern (mel-spectrogram) **Add prosodies** → Include tones, emotion, pauses (make it natural) **Make final audio** → Convert that pattern into real speech you can hear	Produces more natural and intelligible speech than traditional TTS Works across multiple Indian languages Neural TTS significantly improves speech quality and naturalness	Data scarcity for many Indian languages Requires large training datasets. High computational cost	Improve performance for low resource languages Use data augmentation & multilingual learning Develop lightweight models for mobile devices

LITERATURE REVIEW

STUDY	THE METHODOLOGY	KEY FINDINGS	LIMITATIONS	FUTURE SCOPE
Sarvam Saaras Focused on accurate speech-to-text conversion in Indian languages, handling accent diversity and multilingual speech. Focus: real-world Indian speech (noisy, mixed languages).	**Take audio** • Record what you speak **Turn sound into visual pattern** • Convert audio into a spectrogram (like a picture of sound) **Understand the sound** • Neural network learns what the sounds mean **Convert to text** • Output the spoken words as text	High accuracy for India-centric accents and languages Handles code-mixed speech (e.g., Hinglish) Works in noisy real-world environments Suitable for: Voice assistants Transcription systems	Needs large training datasets Performance drops for: Low-resource languages Heavy background noise High computational cost	Improve low-resource language support Better noise robustness Develop on-device lightweight models Integration with TTS for end-to-end voice systems

REFERENCES

[1] World Health Organization,
"Dementia," Sep. 2023. Accessed: Dec. 29, 2025.
[Online]. Available: https://www.who.int/news-room/fact-sheets/detail/dementia

[2] Google,
"Gemini API Documentation."
[Online]. Available: https://ai.google.dev/

[3] SOS Alert System,
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REFERENCES

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[Online]. Available: https://www.mdpi.com/2076-3417/15/17/9390

[5] CareYaya's MedaCareLLM system,
"AI glasses support dementia patients," Nov. 05, 2024.
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[6] MongoDB,
"MongoDB Documentation."
[Online]. Available: Add a little bit of body text

CONCLUSION

This project presents a practical application of Artificial Intelligence in healthcare, aimed at supporting individuals with cognitive impairments. The system enhances patient safety, memory assistance, and independence through real-time location tracking, AI-based face recognition. It also includes intelligent reminder modules for daily routines and important events. By automating monitoring and assistance tasks, the system improves patient independence while significantly reducing caregiver workload, forming a strong foundation for future cognitive assistive technologies.

THANK YOU

For Your Attention

VISION & MISSION OF THE DEPARTMENT

Vision

To evolve as a school of computing with globally reputed Centre's of excellence and serve the changing needs of the industry and society.

Mission

- The department is committed in bringing out career-oriented graduates who are industry ready through innovative practices of teaching and learning process
- To nurture professional approach, leadership qualities and moral values to the graduates by organizing various programs periodically
- To acquire self-sustainability and serve the society through research and consultancy